

Neuro-Symbolic Transformation of Architectural Facades into Procedural Representations

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Figure 1: Cinematic view of a street with facades and their variations all generated by our method.

ABSTRACT

We introduce a neuro-symbolic transformer model that converts flat, segmented facade structures into procedural definitions using a custom-designed split grammar. To facilitate this, we first develop a split grammar tailored for architectural facades and generate a dataset of facades alongside their procedural representations. This dataset is used to train our transformer model to convert segmented, flat facades into the procedural language of our grammar.

CCS CONCEPTS

• **Computing methodologies** → **Shape modeling; Neural networks.**

KEYWORDS

neurosymbolic, procedural facade modeling, transformers

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1 INTRODUCTION

Procedural methods have significantly advanced architectural modeling in computer graphics, enabling efficient generation of complex urban structures [Müller et al. 2006; Wonka et al. 2003]. However, creating grammar rules is time-consuming and requires expertise. To address this, inverse procedural modeling methods extract rules from existing models or images, simplifying the process for non-experts [Musialski and Wimmer 2013; Wu et al. 2014]. This approach also comes with its drawbacks like their complexity, error-proneness, and limited scaling abilities. As a solution, our work introduces a novel neuro-symbolic [Ritchie et al. 2023] approach to enhance facade design. This integration combines neural networks' learning capabilities with procedural models' symbolic reasoning, creating a robust framework for complex architectural modeling. We propose a transformer-based model that converts flat, segmented facade structures into hierarchical procedural definitions using a custom-designed split grammar tailored for architectural facades [Wonka et al. 2003]. Key contributions of our work include: (1) A neuro-symbolic transformer-based model that automates the conversion of segmented facades into editable procedural definitions. (2) A custom-designed split grammar specifically for architectural facades. (3) A novel dataset of facade structure and their procedural definitions, serving as a training and validation foundation. (4) Demonstrations of the model's practical application in generating varied facade designs, showcasing its utility.

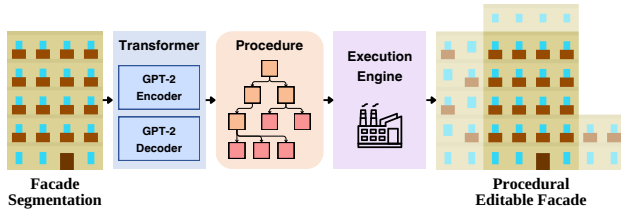


Figure 2: The pipeline of our system: (1) a user provides a segmented facade; (2) a transformer-based model inversely extracts the procedure able to generate the facade; (3) this procedure can be used to regenerate the facade procedurally or to interactively create new variations.

2 OUR APPROACH

2.1 Procedural Data Generation

To our best knowledge, there are no publicly available datasets that contain facades annotated with their procedural definitions of any kind. To create a dataset with facades annotated by procedural definitions, we developed a split grammar system, drawing on facade modeling techniques from previous work [Wonka et al. 2003]. The generation process starts with selecting style-specific arguments and then leverages heuristic modules to select architecturally plausible productions that are consistent with each other. The generated grammar derivation trees, akin to shader graphs but for spatial partitioning, represent the procedural language our model learns.

2.2 Transformer-Based Neuro-symbolic Model

Our transformer model is inspired by recent works which utilize transformers to represent and generate geometric entities, like plants [Lee et al. 2024] or apartment layouts [Leimer et al. 2022]. In contrast to the decoder-only models of previous work, our model is comprised of a transformer-based encoder g_θ which then provides its outputs as context for a transformer-based sequence generator/decoder h_ϕ . We implement g_θ and h_ϕ as GPT-2 models using the implementation included in the Huggingface library.

In order to feed the data through our model, the inputs and outputs need to be converted into sequences of tokens. The input sequence represents a facade segmentation comprised of non-overlapping rectangles of different classes (like *Window*, *Balcony*, etc.). Each rectangle r_i is represented by 5 tokens ($c_i, a_i^x, a_i^y, b_i^x, b_i^y$) where c_i indicates the class of the rectangle and a_i and b_i are the coordinates of its corners. All of the rectangles r_i are then sorted based on the position of their centers and concatenated. Formulating the output procedures poses a greater challenge. Since the procedures take the form of rooted trees we employ breadth-first traversal to produce a sequence of nodes n_i . After that we can resume with a similar approach as with the inputs - each node n_i is represented by its class c_i and the set of its parameters ($p_i^1, \dots, p_i^j$). The nodes n_i are then concatenated with separator tokens in-between them. After tokenization the model is trained on all sequence pairs using cross-entropy loss and teacher forcing.

The neuro-symbolic generation (cf. Figure 2) is performed by providing a segmented facade, which can be done either manually or by using automatic methods [Tyleček and Šára 2013]. From

that flat box-like representation, our model infers and outputs the corresponding procedure (cf. Figure 3, left hand side). Subsequently, the user is free to adjust the parameters of the procedure and create new facade variations (cf. Figure 3, right hand side).

3 RESULTS

We have trained our model on 10,000 procedurally generated pairs of facade segmentations and split grammar based procedures. The model was used to generate all the buildings visible in Figure 1 and Figure 3. Segmentations of target facades were provided as input in the inference process to acquire their procedural representations. After that the procedures were slightly tweaked by the user to get buildings with the same general structure but for instance with more floors, different sizings or alternative element patterns.

REFERENCES

- Jae Joong Lee, Bosheng Li, and Bedrich Benes. 2024. Latent L-Systems: Transformer-Based Tree Generator. *ACM Trans. Graph.* 43, 1, Article 7 (Feb. 2024), 16 pages. <https://doi.org/10.1145/3627101>
- Kurt Leimer, Paul Guerrero, Tomer Weiss, and Przemyslaw Musialski. 2022. LayoutEnhancer: Generating Good Indoor Layouts from Imperfect Data. In *SIGGRAPH Asia 2022 Conference Papers (SA '22)*. Association for Computing Machinery, New York, NY, USA, Article 27, 8 pages. <https://doi.org/10.1145/3550469.3555425>
- Pascal Müller, Peter Wonka, Simon Haegler, Andreas Ulmer, and Luc van Gool. 2006. Procedural modeling of buildings. *ACM Transactions on Graphics* 25, 3 (jul 2006), 614. <https://doi.org/10.1145/1141911.1141931>
- Przemyslaw Musialski and Michael Wimmer. 2013. Inverse-Procedural Methods for Urban Models. In *Proc. of 1st Eurographics Workshop on Urban Data Modelling and Visualisation*, V. Besuevsky and G. Tourre (Eds.). Eurographics Association, Girona, Spain, 31–32. <https://doi.org/10.2312/UDMV/UDMV13/031-032>
- Daniel Ritchie, Paul Guerrero, R. Kenny Jones, Niloy J. Mitra, Adriana Schulz, Karl D. D. Willis, and Jiajun Wu. 2023. Neurosymbolic Models for Computer Graphics. *Computer Graphics Forum* 42, 2 (2023), 545–568. <https://doi.org/10.1111/cgf.14775> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.14775>
- Radim Tyleček and Radim Šára. 2013. Spatial Pattern Templates for Recognition of Objects with Regular Structure. In *Proc. GCPR*. Saarbrücken, Germany.
- Peter Wonka, Michael Wimmer, François Sillion, and William Ribarsky. 2003. Instant architecture. *ACM Transactions on Graphics* 22, 3 (jul 2003), 669. <https://doi.org/10.1145/882262.882324>
- Fuzhang Wu, Dong-Ming Yan, Weiming Dong, Xiaopeng Zhang, and Peter Wonka. 2014. Inverse procedural modeling of facade layouts. *ACM Trans. Graph.* 33, 4, Article 121 (jul 2014), 10 pages. <https://doi.org/10.1145/2601097.2601162>

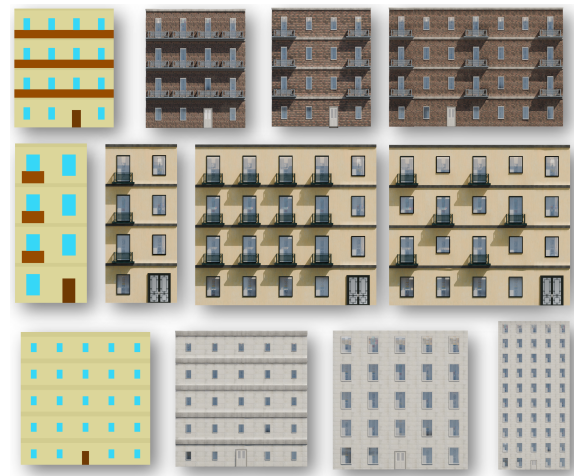


Figure 3: Results of inference and procedure tweaking. From left to right: User Input, Inferred Procedural Representation, Procedural Variation 1, Procedural Variation 2.